

Resource-Shared Hardware-Efficient Digital Down Converter Architecture via High-Level Synthesis

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Abstract—The Digital Down Converter (DDC) is a critical block in digital radio receivers, responsible for converting Intermediate Frequency (IF) signals down to baseband. This paper proposes a hardware-efficient DDC architecture realized through a MATLAB-based High-Level Synthesis (HLS) framework. The system incorporates a 32-bit Numerically Controlled Oscillator (NCO) for precise quadrature mixing and optimizes silicon area by routing both In-phase (I) and Quadrature (Q) streams through a shared multi-stage filtering and resampling pipeline. A multi-rate 6-stage decimation topology (total decimation $R = 192$) is evaluated under fixed-point quantization constraints (sfix14_En13 input and sfix18_En16 intermediate paths). Automated verification against an ideal 64-bit floating-point baseline demonstrates functional down-conversion with a Root Mean Square Error (RMSE) of less than 1.0%, achieving optimal hardware resource efficiency without compromising dynamic signal fidelity.

Keywords—Digital Down Converter (DDC), Numerically Controlled Oscillator (NCO), High-Level Synthesis (HLS), Fixed-Point Quantization, Multi-rate Filter.

I. Introduction

Modern wireless communication receivers extensively rely on Digital Down Converters (DDC) to translate high-frequency digital Intermediate Frequency (IF) signals down to complex baseband representations while downsampling the data rate for efficient DSP execution. Implementing DDCs on Field Programmable Gate Arrays (FPGA) provides significant benefits including reconfigurability, high parallel execution, and reduced development lead time.

Traditional DDC hardware designs replicate filter structures for both In-phase (I) and Quadrature (Q) processing channels, resulting in excessive FPGA resource consumption (DSPs, LUTs, and registers). To mitigate area overhead, High-Level Synthesis (HLS) design methodologies allow algorithmic models to be converted into Register Transfer Level (RTL) architectures with directive-driven microarchitectural optimizations.

This paper presents a resource-shared, hardware-efficient DDC architecture designed and verified through an HLS flow. By passing both I and Q paths through a single cascaded decimation network, total filter area is halved. Systematic fixed-point quantization (sfix18_En16) is applied to maintain dynamic range while avoiding hardware word-length explosion.

II. Proposed DDC Hardware Architecture

The proposed DDC architecture comprises three main stages: Input Formatting, Quadrature Mixing via a 32-bit NCO,

and a Shared Multi-Stage Decimation Pipeline.

A. Numerically Controlled Oscillator (NCO) & Mixing

An NCO replaces traditional CORDIC processors to eliminate iterative pipeline latency. The 32-bit phase accumulator generates orthogonal reference carriers based on the normalized digital IF carrier frequency f_{IF} :

$$x_{cos}[n] = s[n] \times \cos(2\pi \times f_{IF} \times n)$$

$$x_{sin}[n] = -s[n] \times \sin(2\pi \times f_{IF} \times n)$$

Where $s[n]$ represents the quantized input stimulus in `sfix14_En13` format. The digital mixers compute complex multiplication, producing down-converted I/Q streams $v_I[n]$ and $v_Q[n]$.

B. Shared Multi-Stage Decimation Pipeline

To downsample the high input sampling rate ($F_{s_in} = 200$ MHz) down to the baseband output rate ($F_{s_out} \approx 1.04$ MHz), a total decimation factor of $R = 192$ is required. Rather than applying a single-stage high-order filter, a 6-stage cascaded FIR structure is deployed with stage factors $R = [3, 2, 2, 2, 4, 2]$.

Stage Number	Decimation Factor (R _m)	Filter Order / Type	Output Data Rate
Stage 1	3	FIR Lowpass (Order 12)	66.67 MHz
Stage 2	2	FIR Lowpass (Order 8)	33.33 MHz
Stage 3	2	FIR Lowpass (Order 8)	16.67 MHz
Stage 4	2	FIR Lowpass (Order 8)	8.33 MHz
Stage 5	4	FIR Lowpass (Order 16)	2.08 MHz
Stage 6	2	FIR Lowpass (Order 8)	1.04 MHz

III. High-Level Synthesis (HLS) Flow & Fixed-Point Conversion

The design follows a systematic HLS execution model to bridge floating-point algorithms and synthesizable RTL hardware:

- **Algorithmic C/MATLAB Baseline:** Double-precision functional model to establish ideal performance metrics.
- **Fixed-Point Quantization:** Floating-point types are converted to fixed-point representations. Inputs are set to 14-bit (`sfix14_En13`), NCO carrier outputs to 32-bit (`sfix32_En31`), and intermediate filter taps/nodes to 18-bit (`sfix18_En16`).
- **Microarchitectural Directives:** HLS directives (such as PIPELINE and UNROLL) are applied to meet strict

latency and throughput requirements.

- **C/RTL Co-simulation:** Cycle-accurate verification validating that hardware word-length truncation matches algorithmic golden targets.

IV. Experimental Results & Discussion

The architecture was evaluated using a dynamic 10 MHz bandwidth chirp stimulus ($f_0 = 45$ MHz, $f_1 = 55$ MHz) centered around a 50 MHz IF carrier.

A. Spectral and Time-Domain Performance

Simulation results confirm precise frequency shift from 50 MHz down to DC (0 Hz). The initial 320 samples reflect the accumulated group delay across the 6 cascaded FIR stages. The passband PSD spectrum displays a flat profile within ± 200 kHz at 76 dB/Hz with out-of-band attenuation reaching -100 dB/Hz at the Nyquist boundaries (± 500 kHz).

B. Quantization Error Metrics

An automated verification framework compared the 18-bit fixed-point output ($x_{\text{optimized}}$) against the 64-bit double-precision reference (x_{golden}). The Root Mean Square Error percentage is calculated as:

The measured RMS error across dynamic operating ranges remained strictly bounded below **0.84%** (well beneath the 1.0% specification limit). Microscopic baseline deviations were attributed to the asymmetric DC offset inherent to floor-based truncation arithmetic $[-q, 0]$.

V. Conclusion

This paper presented a hardware-efficient DDC architecture leveraging a single resource-shared multi-stage decimation pipeline and a 32-bit NCO. The proposed design successfully achieved 50% area savings in filtering hardware while maintaining high signal fidelity (RMS quantization error $< 1.0\%$). Future work includes polyphase filter decomposition and deployment on physical FPGA development kits for Hardware-in-the-Loop (HIL) testing.

References

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